WEEK 2 INTERNSHIP REPORT

Network Traffic Analysis • Nmap • Windows Security Logging

Organisation: DG Interns Hub

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Summary of learning activities, practical exercises, observations and evidence completed during Week 2.

1. Executive Summary

• During Week 2, I focused on network traffic analysis, network reconnaissance and Windows security event monitoring. I reported completion of six TryHackMe rooms, captured and analysed traffic with Wireshark, scanned an authorised Metasploitable 2 lab VM with Nmap, and investigated Windows events on a Windows 11 laptop.

• The practical work improved my understanding of network protocols, exposed services, Windows audit records and the use of PowerShell for event investigation.

2. Objectives

• Understand TCP connection establishment and termination, including SYN, SYN-ACK, ACK and FIN flags.

• Identify DNS, HTTP and ICMP traffic using Wireshark.

• Use Nmap for host discovery, SYN scanning, service/version detection, OS detection and aggressive scanning against an authorised lab target.

• Investigate Windows Event Viewer records and common Security Event IDs.

• Use PowerShell to query event records and export information to CSV.

3. Lab Environment and Tools

• Kali Linux: 192.168.144.128; Metasploitable 2: 192.168.144.129; Windows environment: personal Windows 11 laptop.

• Tools: Wireshark, Nmap 7.99, Windows Event Viewer, PowerShell and TryHackMe.

• All Nmap scans described in this report were performed against the user's own authorised Metasploitable 2 lab VM.

4. TryHackMe Learning Activities

• Six rooms were reported complete: Nmap; Live Host Discovery; Basic Port Scans; Windows Basics; Windows Fundamentals 1; Windows Fundamentals 2. Individual writeups were prepared for these rooms.

5. Wireshark Practical: Live Traffic Capture

• Captured traffic on Kali's active eth0 interface and generated traffic using ping, nslookup and curl.

• TCP: observed SYN, SYN-ACK and ACK during connection establishment, and FIN/ACK packets during teardown.

• DNS: observed queries and responses for google.com, including A and AAAA records.

• HTTP: observed a HEAD request and 301 Moved Permanently response; a separate GET request was also captured.

• ICMP: observed four echo requests and four replies between Kali (192.168.144.128) and 8.8.8.8.

• Used filters including tcp, dns, http, icmp, tcp.flags.syn == 1 and tcp.flags.syn == 1 && tcp.flags.ack == 0. Saved a PCAPNG capture and screenshots.

6. Nmap Practical: Authorised Lab Scanning

• Connectivity to 192.168.144.129 was verified with four ping replies and no packet loss. Nmap host discovery reported the target as up.

• nmap -sn: host discovery. nmap -sS: SYN scan. nmap -sV: service/version detection. nmap -O: OS detection. nmap -A: aggressive scan.

• The captured SYN scan showed 18 open TCP ports and 982 closed TCP ports.

• Service examples included FTP (21), SSH (22), Telnet (23), HTTP (80), MySQL (3306), PostgreSQL (5432), VNC (5900), IRC (6667), AJP13 (8009) and Tomcat-related HTTP service (8180).

• Service detection identified examples including vsftpd 2.3.4, OpenSSH 4.7p1, Apache httpd 2.2.8 and MySQL 5.0.51a. OS detection estimated Linux 2.6.x.

7. Windows Event Viewer and PowerShell

• Windows logging exercises were performed on a personal Windows 11 laptop because a separate Windows VM was not available.

• 4624 — successful logon; 4625 — failed logon; 4720 — user account created; 4732 — member added to a security-enabled local group; 7045 — service installed (System log).

• Event ID 1102 (Security audit log cleared) was not found in the available history. The Security log was not cleared to manufacture this event.

• Temporary test accounts were created for controlled account-management exercises and removed afterwards.

• PowerShell Get-WinEvent was used to query Event ID 4625 for the previous 30 days, export records to Failed_Logons_4625.csv and group available records by target account.

8. Cheat Sheets and Sample PCAP Analysis

• Prepared a Wireshark display-filter cheat sheet containing 20 filters and a Windows Event ID cheat sheet covering the 15 assigned IDs and logon types 2, 3, 4, 5, 7, 10 and 11.

• dns.cap: observed DNS PTR lookup for 104.9.192.66.in-addr.arpa and A/AAAA lookups for www.netbsd.org.

• http.cap: observed HTTP GET /download.html and an HTTP/1.1 200 OK response.

• ipv4frags.pcap: observed IPv4 fragmentation involving ICMP; Wireshark indicated reassembly in packet number 2 and the visible identification value was 0xbd50.

9. Skills and Learning Outcomes

• Packet analysis and protocol identification with Wireshark; host discovery and service identification with Nmap; interpretation of Windows Security events; PowerShell event retrieval and CSV export; technical documentation and evidence organisation.

10. Challenges and Limitations

• A separate Windows VM was not available, so Windows exercises were performed on the personal Windows 11 laptop.

• Event ID 1102 was not observed and was not forced by clearing the audit log.

• Some additional Nmap text-output files and a separate Nmap PCAP file were not visible in the folder screenshots reviewed during report preparation; add them if available and required.

11. Conclusion

• Week 2 combined network-level and host-level security analysis. Wireshark provided visibility into protocol behaviour, Nmap identified hosts and services in an authorised lab, and Windows Event Viewer and PowerShell supported investigation of logon and account-management activity. These exercises strengthened foundational network security and SOC analysis skills.